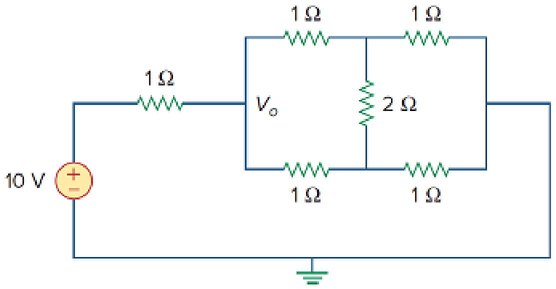
Chapter 2, Problem 72P

Problem

Find *Vo* in the two-way power divider circuit in Fig. 2.132.

Figure 2.132:



Step-by-step solution

Step 1 of 7

(a)

Consider the following circuit diagram:

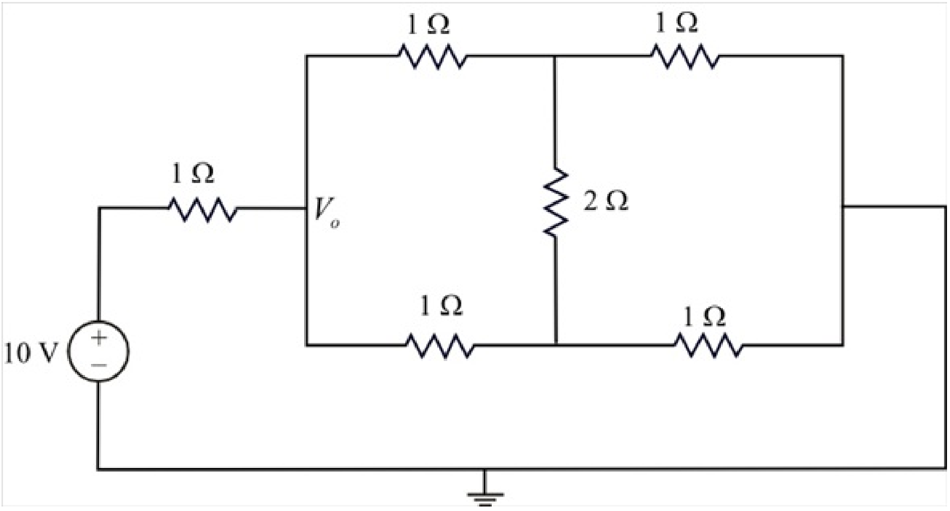


Figure 1

Step 2 of 7

Convert the delta sub network in wye network.

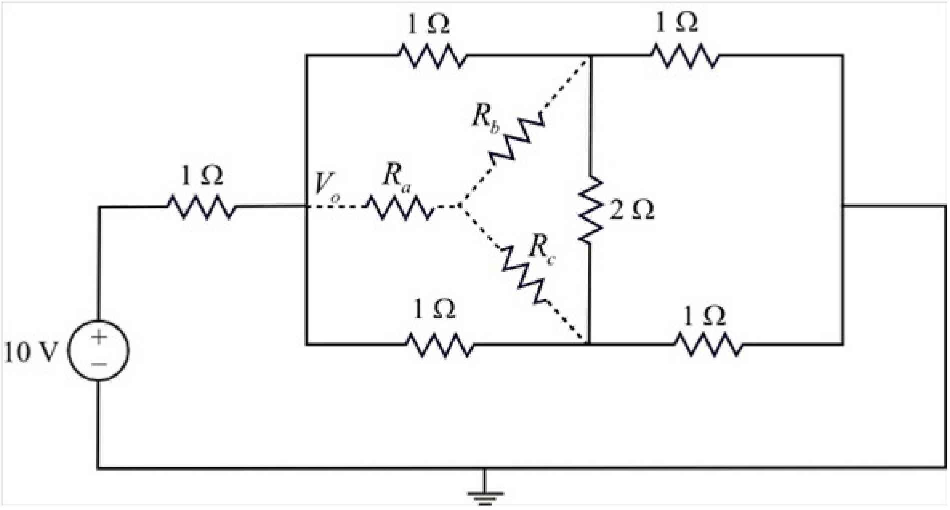


Figure 2

Step 3 of 7

Calculate the star connected resistances.

$$R_a = \frac{(1)(1)}{1+1+2}$$
$$= \frac{1}{4}\ \Omega$$

$$R_b = \frac{(1)(2)}{1+1+2}$$
$$= \frac{2}{4}$$
$$= \frac{1}{2}\ \Omega$$

$$R_c = \frac{(1)(2)}{1+1+2}$$
$$= \frac{2}{4}$$
$$= \frac{1}{2}\ \Omega$$

Step 4 of 7

Consider the modified circuit of Figure 1:

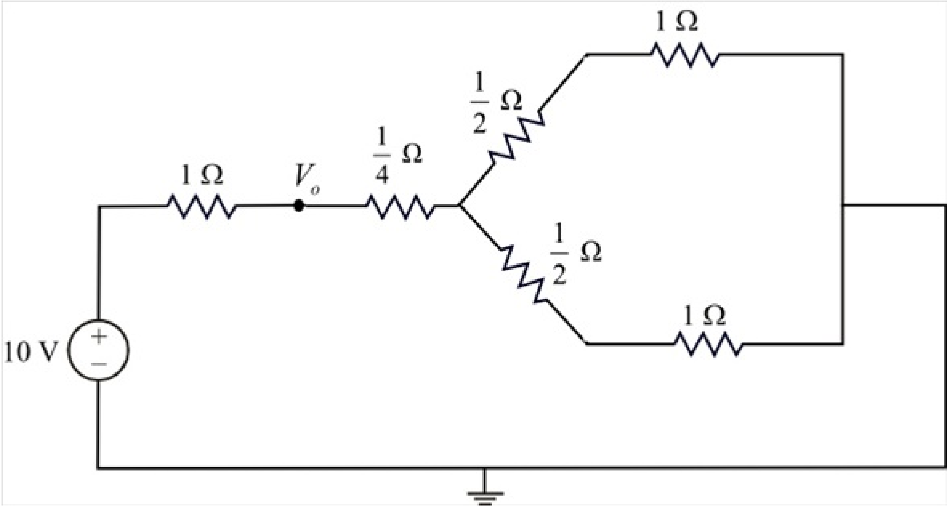


Figure 3

In the Figure 3, the resistors $\frac{1}{2}\,\Omega$ and $1\,\Omega$ are connected in parallel series. Therefore, the resultant equivalent resistance is,

$$\frac{1}{2}+1=\frac{3}{2}\,\Omega$$

Draw the reduced circuit of Figure 3.

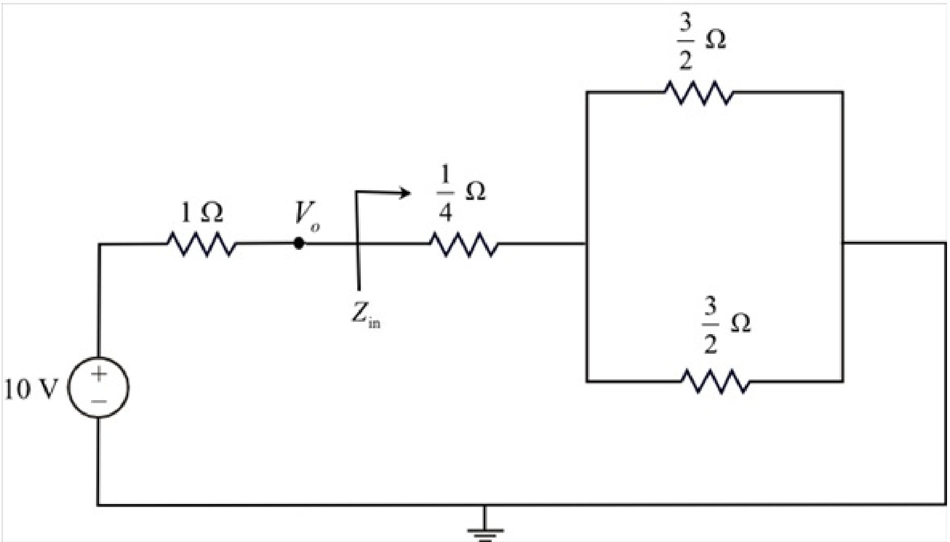


Figure 4

Step 5 of 7

Calculate the impedance Z_{in} in Figure 4.

$$\begin{aligned}
Z_{in} &= \left(\frac{3}{2}\,||\,\frac{3}{2}\right)+\frac{1}{4} \\
&= \frac{\left(\frac{3}{2}\right)\left(\frac{3}{2}\right)}{\frac{3}{2}+\frac{3}{2}}+\frac{1}{4} \\
&= \frac{3}{4}+\frac{1}{4} \\
&= 1\,\Omega
\end{aligned}$$

Step 6 of 7

Draw the reduced circuit of Figure 4.

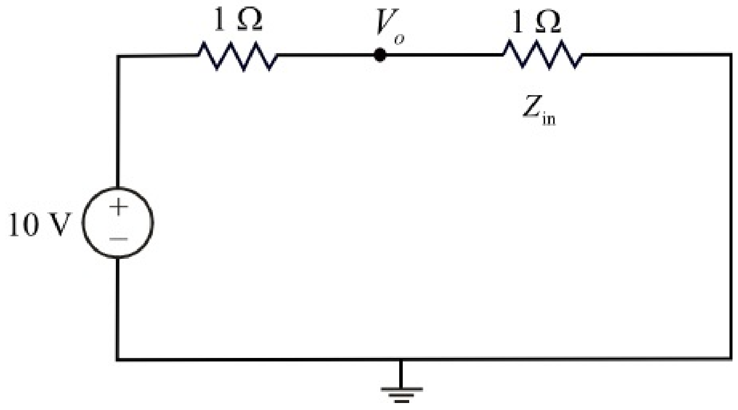


Figure 5

Step 7 of 7

Calculate the voltage V_o in Figure 5 using voltage divider rule.

$$\begin{aligned}V_o &= (10)\left(\frac{1}{1+1}\right)\\ &= (10)\left(\frac{1}{2}\right)\\ &= 5\text{ V}\end{aligned}$$

Therefore, the voltage V_o is $\boxed{5\text{ V}}$.